

Lab Assignment 2

What:

In this assignment you are to write a C or Fortran program that sends a token around the network for an arbitrary number of nodes:

1. Login to prospero via ssh (`ssh yourusername@prospero.wit.edu`)
2. Write a Slurm script to submit to the supercomputer.
3. Use MPI to send a token sequentially around the network for **any number of nodes**.
4. Compute the network bandwidth using two communication patterns: ring and ping-pong.
5. Verify that your code is working correctly.
6. Create a **PDF** file with a write-up about this assignment. Leave the pdf file in the same directory where your assignment is.

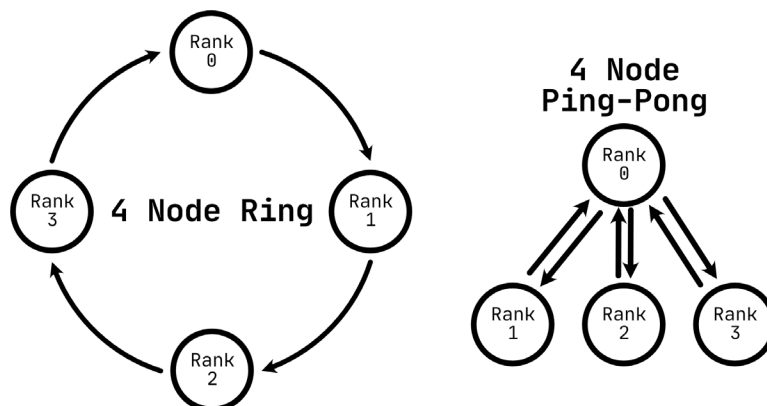
Why:

Understanding how to send and receive data to/from other nodes is imperative in this class. While you may not always use Send/Recv pairs in all codes that we write, these MPI calls are the basis for the message-passing paradigm. In addition, you will be timing the execution, which is needed in future assignments to examine the performance of your parallel code.

How:

Start by sending a single integer between two nodes, like we did in class, and confirm that your program is working correctly. For the ring, you do not need any loops, only an if/else style structure. Remember that each rank is running the exact same code at the exact same time; use the if/else to direct the ranks to run the appropriate code. The ping-pong style communication will require a loop, but only for rank 0.

Now, try sending a larger structure, like an array, and time how long it takes to travel between the nodes. Take note of where you put your timers, as putting them in different places will give you different information. The size of the array should be large, ~1GB. Use this information to compute the bandwidth of the network (GB/s and/or Gb/s).



Expected Result:

Describe the differences between the two algorithms, how they create their respective communication pattern, and confirm that both report the same network bandwidth ($\pm$ a small amount based on network traffic). The bandwidth is not dependent on the communication pattern. Include the bandwidth for both patterns in your writeup. Our hardware claims to be 10Gb/s (or 1.25GB/s), you should get a value close to this.